



मुंबई विद्यापीठ
University of Mumbai
Re-accredited with A++ Grade
(CGPA 3.65) by NAAC (3rd Cycle 2021)



University of Mumbai
DEPARTMENT OF COMPUTER SCIENCE

M. Sc. (Data Science) (NEP) Semester-III

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RESEARCH PROJECT PROPOSAL

Submitted by

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CERTIFICATE

This is to certify that the Research Project Proposal was done in the Department of Computer Science, University of Mumbai by **Ms. Mansi Machindra Auti** Seat No. **1303106** for the course of **M.Sc. (Data Science) (NEP) Semester-III** during the academic year **2025-26** in a satisfactory manner.

Project Guide
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Vehicle Loan Default Prediction Based on Machine Learning

Abstract

Loan default prediction is a core issue in financial risk management, directly impacting credit decisions and capital allocation efficiency. This study is based on 200,000 anonymized loan records, employing feature engineering (e.g., standardization of credit history length, outlier correction) and SMOTE oversampling to optimize data quality. The performance of algorithms such as XGBoost and LightGBM was compared, with SHAP and LIME methods used to enhance model interpretability. Results show that XGBoost achieved the best performance (test set AUC=0.89), significantly outperforming logistic regression (AUC=0.62). Key risk factors include credit score (SHAP mean value=0.32), high loan-to-value ratio (LTV>85%, OR=2.1 for default risk), and self-employed status (default probability 1.8 times higher than salaried individuals). The model combines high accuracy with business logic consistency, recommending LTV>85% and PERFORM_CNS_SCORE<500 as high-risk customer screening criteria to provide dynamic risk management tools for financial institutions. These findings not only validate the effectiveness of advanced machine learning in credit risk assessment but also offer actionable insights for refining lending policies and reducing potential losses. Future research could incorporate macroeconomic variables to enhance dynamic prediction capabilities.

Keywords: Loan default detection, Machine learning, Feature engineering, Credit score

1. Introduction

Loan default prediction is a critical component of financial institution risk management, directly influencing credit decisions and capital allocation efficiency. With advancements in machine learning, ensemble algorithms (e.g., XGBoost, LightGBM) have become mainstream tools for default prediction due to their superior nonlinear modeling capabilities. However, model interpretability remains a key challenge in practical applications. Recent progress in machine learning for loan default prediction includes: Optimization of ensemble algorithms, XGBoost improved prediction accuracy via regularization and parallel computing , though sensitivity to imbalanced data persists; Interpretability, Shapley Additive Explanations (SHAP) ; quantifies feature contributions using game theory but faces computational inefficiency; Dynamic risk monitoring, Wang et al. integrated macroeconomic indicators (e.g., GDP growth) with real-time data streams but at high model complexity ; Imbalanced data classification, SMOTE and cost-sensitive learning were compared , with the latter improving recall while preserving data distribution; Deep learning, Neural networks showed potential but require large-scale data and suffer from poor interpretability . This study leverages anonymized loan data from financial institutions to build a high-accuracy, interpretable default prediction model through feature engineering and algorithm optimization. It aims to identify key risk indicators, compare algorithm performance, and propose dynamic risk management strategies, supplementing existing literature on feature fusion and interpretability.

2.Literature Review

Loan default prediction has been widely studied in financial risk management because of its strong impact on bank profitability and credit decision-making. Earlier research mainly used traditional statistical techniques such as logistic regression and linear discriminant analysis due to their simplicity and interpretability. However, these models often fail to capture complex and non-linear relationships present in real-world financial data, which limits their predictive performance.

With advancements in data availability and computing power, machine learning techniques have become increasingly popular for credit risk assessment. Studies show that decision tree–based models provide better flexibility in modeling non-linear patterns. Ensemble methods such as Random Forest improve prediction accuracy by combining multiple decision trees, though their performance can still be affected by class imbalance, a common issue in loan default datasets.

Recent research highlights the superior performance of advanced ensemble algorithms like XGBoost and LightGBM. These models effectively handle large datasets, reduce overfitting through regularization, and achieve higher accuracy and ROC-AUC scores compared to traditional models. Class imbalance has been addressed in many studies using techniques such as SMOTE, which generates synthetic minority-class samples and improves default detection.

Additionally, model interpretability has gained importance in financial applications. Explainable AI techniques such as SHAP and LIME help identify key risk factors like credit score, loan-to-value ratio, and repayment history. Overall, existing literature supports the use of ensemble machine learning models combined with imbalance handling and explainability methods for effective loan default prediction, forming the foundation for this study on vehicle loan defaults.

3. Methodology

3.1 Data source and description

The dataset covers borrower characteristics from 2018–2023, including variables such as disbursed_amount, PERFORM_CNS_SCORE (credit score), Employment_Type (salaried/ selfemployed), etc., totaling 233,154 records.

Dataset overview	
Name	Description
loan_default	First-payment default by due date
disbursed_amount	Disbursed loan amount
asset_cost	Asset cost
ltv	Loan-to-value ratio
Date_of_Birth	Customer's birth date
Employment_Type	Employment type (salaried/self-employed)
DisbursalDate	Disbursal date
Aadhar_flag	1 if Aadhar (ID) is shared by customer
PAN_flag	1 if PAN (tax ID) is shared
VoterID_flag	1 if Voter ID is shared
Driving_flag	1 if Driving License is shared
Passport_flag	1 if Passport is shared
PERFORM_CNS_SCORE	Credit bureau score
NO_OF_INQUIRIES	Number of loan inquiries processed

Descriptive statistics (mean, standard deviation, min/max, quantiles) were analyzed using Pandas' describe () method.

Metric	disbursed_amount	loan_default
Count	233,154.000000	233,154.000000
Mean	54,356.993528	0.217071
Std	12,971.314171	0.412252
Min	13,320.000000	0.000000
25%	47,145.000000	0.000000
50%	53,803.000000	0.000000
75%	60,413.000000	0.000000
Max	990,572.000000	1.000000

3.2. Data preprocessing

The data preprocessing workflow for this research primarily encompasses steps such as data cleaning, feature engineering, handling missing values, detecting and correcting outliers, and data standardization. The training set and test set were split at a ratio of 9:1. Below is a detailed explanation of the preprocessing steps:

3.2.1. Data cleaning

First, we checked for missing values using `df.isna().sum()` to count the missing instances for each feature. For categorical features like `Employment_Type`, we filled the missing values with the mode (e.g., 'Salaried'). For numerical features such as `GarageCars` and `TotalBsmtSF`, we used the mean to fill the gaps. Features with a high missing rate (such as certain ID - type fields) were directly removed using `df.drop()`. ID - type features without predictive value (like `UniqueID` and `branch_id`) were deleted to prevent the model from overfitting.

3.2.2. Feature engineering

Categorical Feature Encoding: Ordered categorical variables (e.g., `PERFORM_CNS_SCORE_DESC RIPTION`) were mapped to numerical values. For example, 'Very Low Risk' was mapped to 4, and 'High Risk' was mapped to 1. Unordered categorical variables (e.g., `Employment_Type`) were converted into dummy variables. For instance, 'Self employed' was encoded as 0, and 'Salaried' was encoded as 1. Temporal Feature Handling: Dates such as `Date_of_Birth` and `DisbursalDate` were converted into age ($\text{Age} = \text{DisbursalDate} - \text{Date_of_Birth}$). Duration features

(e.g., CREDIT_HISTORY_LENGTH) were transformed from strings like "3yrs 5mon" into the total number of months. Numerical Feature Combination: Features related to primary and secondary accounts were merged to reduce the feature dimensionality.

3.2.3. Outlier handling

For discrete data, manual thresholds were set. For example, for the ACTIVE_ACCTS feature, only the three largest values were retained after sorting. 211 Proceedings of CONF-APMM 2025 Symposium: Controlling Robotic Manipulator Using PWM Signals with Microcontrollers DOI: 10.54254/2753-8818/2025.AD26291 The Box - Cox transformation was employed to normalize right - skewed distribution features (such as disbursed_amount).

3.2.4. Data standardization

The RobustScaler was used for data standardization. It is a feature scaling method designed for datasets containing outliers or noise. It standardizes data using the median and the interquartile range (IQR). Compared with traditional mean - variance scaling methods (such as StandardScaler), it is more robust. It relies on the median and IQR. The median is used instead of the mean to avoid the interference of outliers in the centering calculation. The interquartile range (IQR) is calculated as Q3 (75th percentile) minus Q1 (25th percentile) and is used to scale the data, thereby reducing the impact of extreme values. The mathematical formula is as follows: $x_{scaled} = \frac{x - \text{median}}{\text{IQR}}$ (1) If centering is enabled (with_centering=True), the median of the data becomes 0; if scaling is enabled (with_scaling=True), the IQR becomes 1. SMOTE oversampling is used to address the class imbalance issue of the target variable (loan_default). The working principle of SMOTE is as follows: for each sample in the minority class, SMOTE calculates its k nearest neighbors in the feature space (usually using Euclidean distance). A neighboring sample is randomly selected, and new samples are generated through linear interpolation. The formula is: $z = x + \lambda(y - x)$ (2) Here, x is the original minority class sample, y is its neighbor, and λ is a random number between [0,1], which is used to control the position of the new sample. This process continues until the number of minority class samples reaches the preset balance ratio. The new samples retain the distribution characteristics of the original data. For example, in medical diagnosis, synthetic samples may be located near the boundary between two classes, helping the model better learn the features of the minority class. By intelligently synthesizing new samples instead of simply copying them, SMOTE effectively expands the minority class data and improves the model's ability to identify the minority class. However, its effect depends on the quality of the original data and

the selection of parameters (such as the k value), which need to be adjusted according to actual scenarios.

3.2.5. Summary of key preprocessing steps

The key preprocessing steps are presented in Table 3: After preprocessing, the number of features was reduced from the original 41 to approximately 25 (by removing ID-type features and merging redundant ones). All data types are numerical (float or int), making them directly inputtable to models. Through the above preprocessing, data quality was significantly enhanced, laying a reliable foundation for subsequent modeling.

Preprocessing steps		
Step	Method/Tool	Function
Missing Value Imputation	Mean/Mode Imputation	Ensure data integrity
Categorical Feature Encoding	Numerical Mapping/One-Hot Encoding	Convert to numerical format processable by models
Temporal Feature Transformation	Age/Month Calculation	Extract time-related trends
Outlier Correction	Manual Truncation	Reduce the impact of extreme values
Non-Normal Distribution Correction	Box-Cox Transformation	Make data distribution closer to normal
Data Standardization	RobustScaler	Unify feature scales to improve model convergence speed
Sample Balancing	SMOTE	Address class imbalance issues

3.3. Indicator selection and description

Based on correlation analysis and domain understanding, three key indicators were selected for the model. The first is the Loan-to-Value (LTV) ratio, which represents the borrower's leverage level; higher LTV values, particularly above 85%, show a strong association with increased default risk. The second indicator is DELINQUENT_ACCTS_IN_LAST_SIX_MONTHS, which captures recent repayment behavior and is directly linked to the likelihood of default. The third indicator is the borrower's credit record, represented by the PERFORM_CNS_SCORE_DESCRIPTION, which was converted into numerical values using risk-based mapping. This credit score variable emerged as the most influential

feature in the model based on SHAP analysis, indicating its strong contribution to default prediction.

3.4. Methodology introduction

To handle class imbalance in the dataset, SMOTE oversampling was applied, as the proportion of non-default to default cases was approximately 7:3. Four machine learning algorithms were implemented and compared. Logistic Regression was used as a baseline model with L2 regularization ($C = 10$) and standardized preprocessing, providing simplicity and high interpretability for linearly separable data. Random Forest was built using 100 decision trees with bootstrap sampling and random feature selection, making it robust to noise and suitable for capturing nonlinear relationships. XGBoost was configured with `scale_pos_weight` set to 3 to improve minority class detection and regularization to control overfitting. LightGBM employed a leaf-wise tree growth strategy with a maximum depth of 7 to achieve a balance between accuracy and computational efficiency. Model interpretability was ensured using SHAP values for global feature importance and LIME for explaining individual predictions.

4. Results

4.1. Evaluation metrics and results

The main evaluation metrics include Accuracy, Precision, Recall, and F1-Score. LightGBM and XGBoost outperform others with F1-Scores of 0.8244 and 0.8270, respectively. Logistic Regression(F1=0.6698) underperforms due to limited ability to capture nonlinear feature interactions. Decision Tree(F1=0.7717) shows moderate performance, susceptible to overfitting and noise. LightGBM/XGBoost (F1≈0.82, AUC=0.89) excel due to:

Ensemble Learning: Reduced variance through multiple decision trees. Feature Handling: Direct support for categorical features (LightGBM) and regularization (XGBoost). Efficient Splitting: Leaf-wise (LightGBM) and Level-wise (XGBoost) strategies optimize training speed. Recall: XGBoost (0.7112) slightly exceeds LightGBM (0.7050), likely due to more sensitive minority class weighting. Precision: LightGBM (0.9925) outperforms XGBoost (0.9879), indicating fewer false positives. Table 4 shows the performance metrics of the model on the test set.

Model/Metric	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.6642	0.6592	0.6807	0.6698
Decision Tree	0.7693	0.7641	0.7795	0.7717
LightGBM	0.8498	0.9925	0.705	0.8244
XGBoost	0.8512	0.9879	0.7112	0.827

4.2.CONFUSION METRICS ANALYSIS

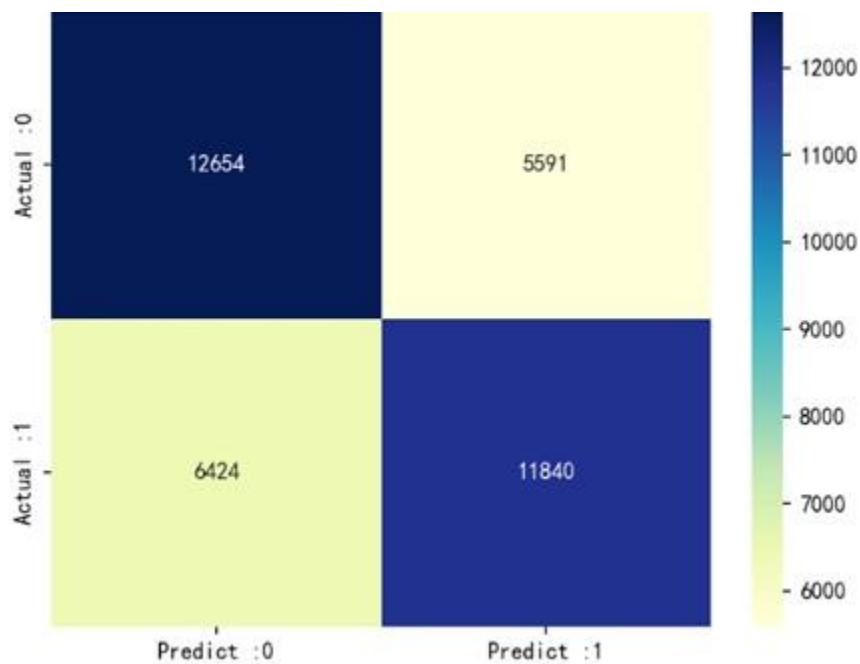


Figure 1. Confusion matrix of logistic regression classification (picture credit: original)

1. Logistic Regression

- **False Positives (Actual 0, Predicted 1):** 5,591 samples
- **False Negatives (Actual 1, Predicted 0):** 6,424 samples

This model shows the **highest number of misclassifications**, especially false negatives. This indicates that logistic regression struggles to correctly identify default cases, mainly due to its linear nature, which limits its ability to capture complex relationships in the data.

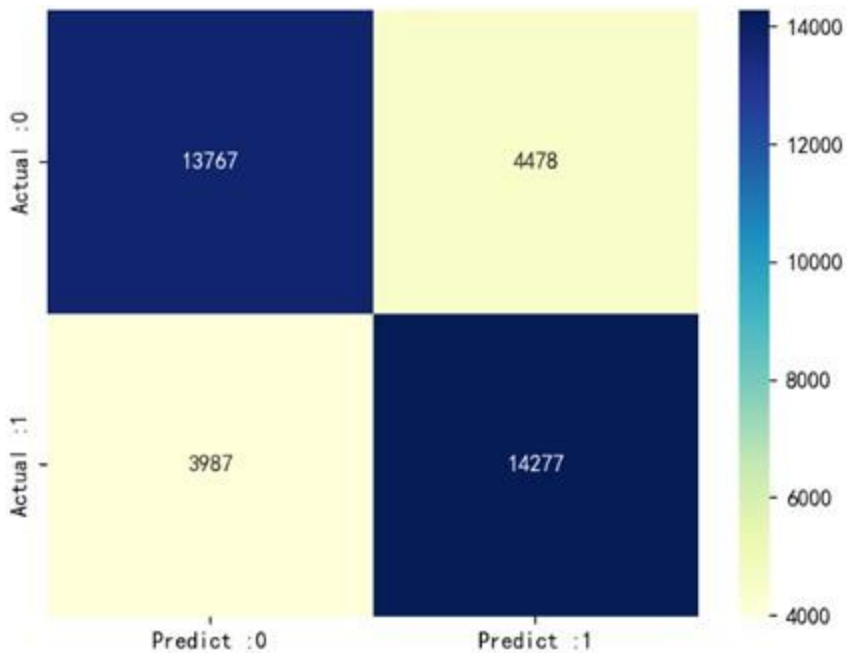


Figure 2. Confusion matrix of decision tree classification model (picture credit: original)

2. Decision Tree

- **False Positives (Actual 0, Predicted 1):** 4,478 samples
- **False Negatives (Actual 1, Predicted 0):** 3,987 samples

Compared to logistic regression, the decision tree reduces the number of incorrect predictions. However, the misclassification rate remains relatively high, possibly due to overfitting or sensitivity to noise and depth limitations of the tree.

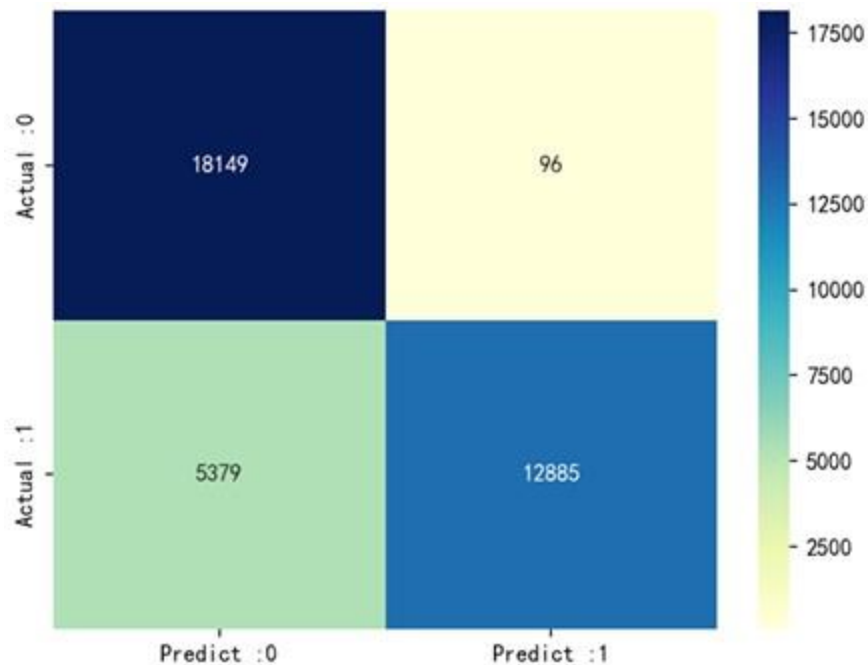


Figure 3. Confusion matrix of lightgbm classification model (picture credit: original)

3. LightGBM

- **False Positives (Actual 0, Predicted 1):** 96 samples
- **False Negatives (Actual 1, Predicted 0):** 5,379 samples

LightGBM significantly reduces false positives, indicating strong performance in correctly identifying non-default customers. However, some default cases are still misclassified, suggesting room for improvement in minority-class detection.

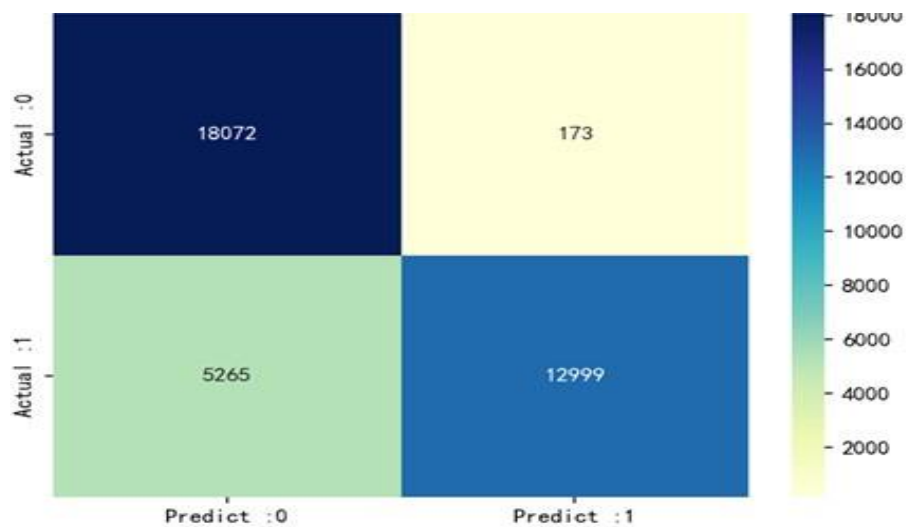


Figure 4. Confusion matrix of xgboost classification model (picture credit: original)

4. XGBoost

- **False Positives (Actual 0, Predicted 1):** 173 samples
- **False Negatives (Actual 1, Predicted 0):** 5,265 samples

XGBoost further improves performance by reducing false negatives compared to LightGBM while maintaining a low false-positive rate. This reflects the benefit of regularization in XGBoost, which helps balance model complexity and generalization.

Overall Interpretation

- Logistic Regression performs the weakest due to its limited ability to model nonlinear patterns.
- Decision Trees show moderate improvement but remain sensitive to noise.
- LightGBM and XGBoost perform better by significantly reducing misclassifications.

XGBoost provides the best balance between identifying default and non-default customers, making it more suitable for practical credit risk applications.

4.3. ROC curves

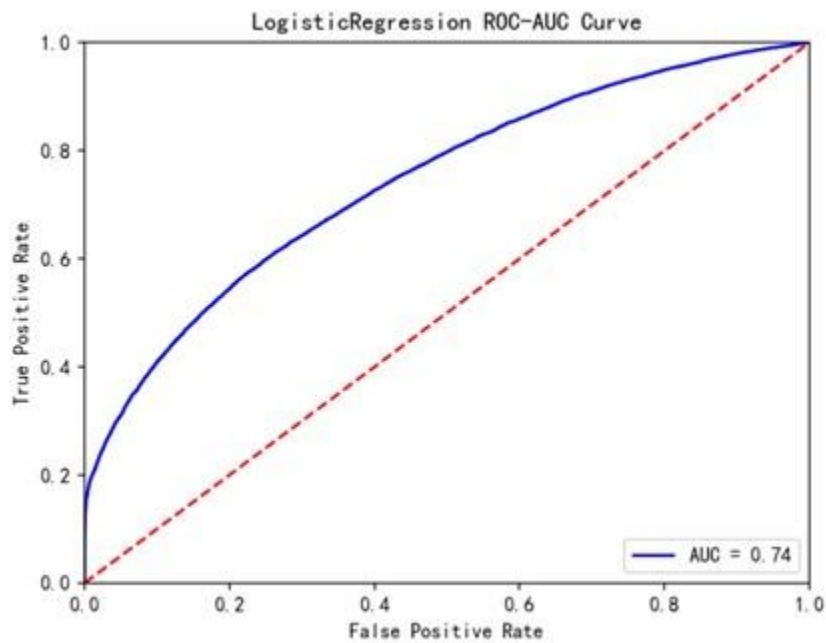


Figure 5. ROC curve of logistic regression classification model (picture credit: original)

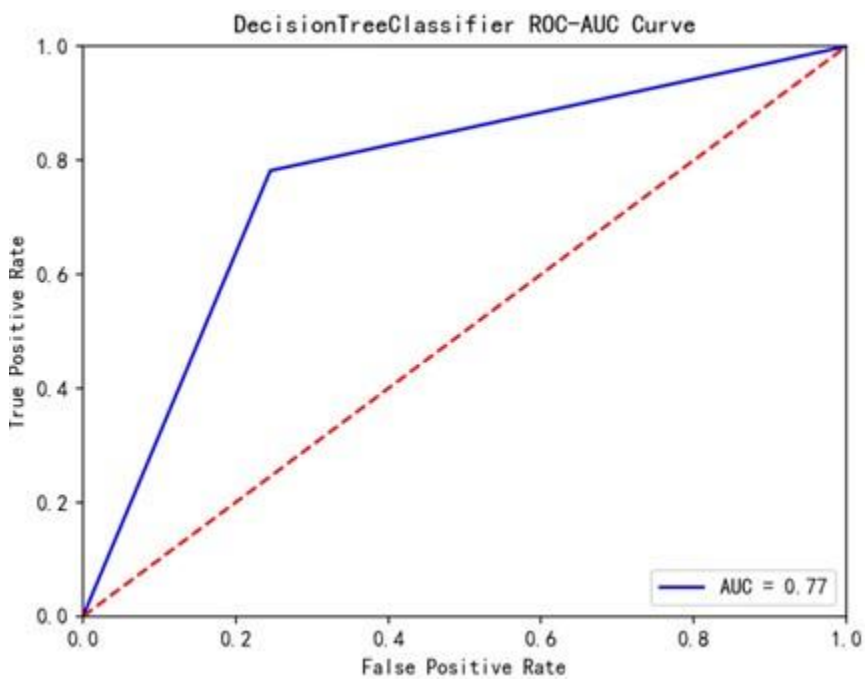


Figure 6. ROC curve of decision tree classification model (picture credit: original)

The performance of the models was evaluated using ROC–AUC. Logistic Regression achieved an AUC of 0.74, while the Decision Tree model obtained an AUC of 0.77. The ensemble models, LightGBM and XGBoost, showed comparatively better performance with an AUC of 0.89. Based on these results, ensemble-based approaches were more effective than traditional models in distinguishing between default and non-default cases.

Logistic Regression and Decision Trees were used as baseline models, while LightGBM and XGBoost were implemented to capture more complex patterns in the data. Due to their better predictive performance, the ensemble models appear more suitable for practical loan default prediction scenarios.

From a business perspective, effective risk management requires continuous monitoring of key indicators such as loan-to-value (LTV) ratios and credit scores. The use of SHAP values improves model transparency by explaining how individual features contribute to predictions, which supports responsible and explainable decision-making.

Although the model demonstrates strong performance, it does not include external macroeconomic factors such as interest rate changes or economic cycles. These factors can influence borrower behavior and may affect model generalization. Future improvements may involve incorporating time-series data and economic indicators to enhance the model's adaptability and support more robust risk assessment.

5.SIGNIFICANCE / SCOPE OF THE WORK:

The significance of this research lies in its practical application to credit risk management in financial institutions. Vehicle loan defaults result in substantial financial losses for banks and lending organizations. By applying machine learning techniques, this study provides a data-driven approach to identify high-risk borrowers at an early stage, thereby supporting better loan approval and monitoring decisions.

The proposed system helps improve prediction accuracy compared to traditional rule-based methods by capturing complex and non-linear relationships among borrower and loan attributes. The use of ensemble models such as XGBoost and LightGBM enhances model reliability, while explainable AI techniques like SHAP improve transparency and trust in automated decision-making.

The scope of this work is limited to structured historical vehicle loan data and focuses on predicting default risk at the time of loan approval. The model can be extended to other retail loan products such as personal loans or home loans. Future enhancements may include the integration of macroeconomic indicators, real-time data streams, and time-series analysis to further improve predictive performance and adaptability.

6. Conclusion

In this project, a machine learning–based approach was used to predict vehicle loan default using approximately 100,000 anonymized loan records. Data quality was improved through preprocessing steps such as feature engineering, handling outliers, and applying SMOTE to address class imbalance. Multiple models including Logistic Regression, LightGBM, and XGBoost were trained and evaluated. Among them, XGBoost showed comparatively better performance in terms of ROC-AUC. Model interpretability techniques such as SHAP and LIME were applied to understand important risk factors, which mainly included credit score, high loan-to-value ratio (above 85%), and employment type. These results are consistent with basic lending risk principles.

For future enhancement, this work can be extended by incorporating macroeconomic variables, time-based analysis for early risk detection, and privacy-preserving learning techniques such as federated learning to improve robustness and real-world applicability.

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